This is also $8x^{-4}$ and $(8x^{-4})' = -32x^{-5}$

In all quizzes and exams, show your work and write with correct notation. To earn points on this group quiz, you MUST participate with your group. The highest score among all groups will be the score applied to everyone participating in ANY group. This quiz is open notes, but no external resources (Internet, ChatGPT, etc.) are allowed. Names and emails must be written on the reverse side.

(30pts) 1. If $f(x) = \sin\left(\sin\left(\sin\left(\sin(5^{\cos x})\right)\right)\right) + \dfrac{3^{4^x+x^7}+x}{\sqrt{x^8+\tan x}} + \csc(6^x)\cdot\dfrac{8}{x^4} - x^4\sin(x)$.

This is also $(x^8+\tan x)^{1/2}$

$$\cos(\sin(\sin(\sin(5^{\cos x}))))\cdot\cos(\sin(\sin(5^{\cos x})))\cdot\cos(\sin(5^{\cos x}))\cdot\cos(5^{\cos x})\cdot 5^{\cos x}\ln 5\cdot(-\sin x)$$

$$+\frac{\left(\sqrt{x^8+\tan x}\right)\left(3^{4^x+x^7}\ln 3\cdot(4^x\ln 4+7x^6)+1\right)-\left(3^{4^x+x^7}+x\right)\left(\frac{1}{2}(x^8+\tan x)^{-1/2}\cdot(8x^7+\sec^2 x)\right)}{\left(\sqrt{x^8+\tan x}\right)^2}$$

could also have $x^8+\tan x$ (the square and $\sqrt{\ }$ undo each other)

$$+\csc(6^x)\cdot 8(-4)x^{-5} + \frac{8}{x^4}\cdot-\csc(6^x)\cot(6^x)\cdot 6^x\ln 6$$

could have $8x^{-4}$ instead

could have $\dfrac{(x^4)(0)-(8)(4x^3)}{(x^4)^2}$ instead

$$-\left(x^4\cos x + \sin x\cdot 4x^3\right)$$

parentheses needed here because of minus sign

Identifying information for EACH participant should be written on THIS side only. (One submission per group.)

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